

# Rewilding the Ledger: A Network-Theoretic Account of Resilience in Urban Food-Sharing Systems

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**Abstract.** Urban food-sharing networks – community fridges, gleaning circuits, and informal surplus exchanges – are usually studied from a single disciplinary vantage: as supply chains, as social capital, or as ecological flows. We argue that their resilience to shocks is properly a joint property of network topology and reciprocity norms, and cannot be recovered from either lens alone. We build an agent-based model of 42 simulated neighborhood clusters in which household nodes exchange surplus food along edges weighted by both physical proximity and a behavioral trust parameter drawn from repeated-game reciprocity theory. Across 1,000 simulated disruption events, empirically-calibrated networks absorbed 31% more shock volume than random-topology baselines of equal density, and recovery time correlated more strongly with redundancy index than with raw connectivity. We propose a composite resilience metric,  $R_i$ , that outperforms degree centrality alone as a predictor of post-shock recovery. The results suggest that interventions aimed at food-system resilience should target trust-mediated redundancy, not merely network density.

**Keywords:** network resilience, food systems, behavioral economics, urban ecology, agent-based modeling



## 42 Clusters

Simulated neighborhoods, 6–38 households each, calibrated against survey data on informal exchange.



## 1,000 Shocks

Monte Carlo disruption events used to estimate recovery-time distributions per cluster.



## Composite $R_i$

A resilience index blending topology and reciprocity that beats degree centrality alone.

## 1 Introduction

Interdisciplinary work on urban resilience has a recurring failure mode: each discipline studies the piece of the system it already has instruments for. Network scientists trace the topology of exchange without asking why an edge exists in the first place; behavioral economists model reciprocity and trust in the laboratory without asking how those norms propagate across a real settlement pattern. Urban food-sharing systems – the loose lattice of community fridges, gleaning routes, and neighbor-to-neighbor surplus exchange that has grown rapidly since 2019 – sit exactly at the seam between these two traditions, and we think they are a useful test case for whether the seam matters.

Our claim is narrow but, we think, underappreciated: the *topology* of a food-sharing network and the *reciprocity norms* riding on top of it are not separable inputs to resilience. A dense network with weak trust degrades under stress in qualitatively different ways than a sparse network with strong trust, and neither a pure graph-theoretic model nor a pure behavioral model predicts which. Section 2 situates this claim against network-science and behavioral-economics treatments of food systems. Section 3 describes an agent-based model that couples the two layers explicitly.

Section 4 reports shock-absorption and recovery outcomes across 1,000 simulated disruptions, and Section 5 draws out what this implies for intervention design.

## 2 Related Work

### 2.1 Network approaches to food systems

Graph-theoretic treatments of food distribution typically borrow directly from supply-chain resilience literature, modeling exchange networks as static graphs and asking how robust they are to node or edge removal. This tradition is good at identifying structurally critical nodes but treats every edge as equally likely to be used once a disruption forces rerouting – an assumption that behavioral evidence does not support.

### 2.2 Behavioral economics of reciprocity

A parallel literature, largely lab-based, models informal exchange as a repeated game in which reciprocity norms sustain cooperation without formal contracts. This work is precise about *when* an actor will honor an exchange relationship under stress, but rarely embeds that finding in a realistic network of who is even reachable to exchange with.